

# Vector Spaces

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# Vector Spaces

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## Definition

A **vector space** is a nonempty set  $V$  of objects, called *vectors*, on which are defined two operations, called *addition* and *multiplication by scalars* (real numbers), subject to the ten axioms (or rules) listed below. The axioms must hold for all vectors  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  in  $V$  and for all scalars  $c$  and  $d$ .

- 1 The sum of  $\mathbf{u}$  and  $\mathbf{v}$ , denoted by  $\mathbf{u} + \mathbf{v}$ , is in  $V$ .
- 2  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ .
- 3  $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ .
- 4 There is a **zero** vector  $\mathbf{0}$  in  $V$  such that  $\mathbf{u} + \mathbf{0} = \mathbf{u}$ .
- 5 For each  $\mathbf{u}$  in  $V$ , there is a vector  $-\mathbf{u}$  in  $V$  such that  $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ .
- 6 The scalar multiple of  $\mathbf{u}$  by  $c$ , denoted by  $c\mathbf{u}$ , is in  $V$ .
- 7  $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ .
- 8  $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ .
- 9  $c(d\mathbf{u}) = (cd)\mathbf{u}$ .
- 10  $1\mathbf{u} = \mathbf{u}$ .



## Properties

Let  $V$  be a vector space, for each  $u \in V$  and scalar  $c$

- 1 There is a single zero element in an arbitrary vector space
- 2 There is a single additive inverse vector for every vector  $u$
- 3  $0u = 0$
- 4  $c0 = 0$
- 5  $-u = (-1)u$



## Example 1. The zero vector space $V = \{0\}$

$$0+0=0$$

$$k0=0$$



## Example 2. The vector space $\mathbb{R}^n$

- $V_1 = \mathbb{R}^n$

$$\mathbf{u} + \mathbf{v} = (u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

$$k\mathbf{u} = k(u_1, u_2, \dots, u_n) = (ku_1, ku_2, \dots, ku_n)$$



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$$k\mathbf{u} = k(u_1, u_2, \dots, u_n) = (ku_1, ku_2, \dots, ku_n)$$

- $V_2 = \{(x_1, x_2, x_3), x_i \in \mathbb{R} \text{ and } 2x_1 + x_2 - x_3 = 0\} \subseteq \mathbb{R}^3$

check:

- 1 The zero vector is in the set
- 2 It is closed under addition
- 3 It is closed under scalar multiplication



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$$k\mathbf{u} = k(u_1, u_2, \dots, u_n) = (ku_1, ku_2, \dots, ku_n)$$

- $V_2 = \{(x_1, x_2, x_3), x_i \in \mathbb{R} \text{ and } 2x_1 + x_2 - x_3 = 0\} \subseteq \mathbb{R}^3$

check:

- ① The zero vector is in the set
- ② It is closed under addition
- ③ It is closed under scalar multiplication

- $V_3 = \{(x_1, x_2, x_3), x_i \in \mathbb{R} \text{ and } x_1 + x_2 - 2x_3 = 1\} \subseteq \mathbb{R}^3$

$$\mathbf{x} = (1, 2, 1) \in V_3, \mathbf{y} = (2, 3, 2) \in V_3, \text{ however, } \mathbf{x} + \mathbf{y} = (3, 5, 3) \notin V_3$$





### Example 3.

$$V = \{(x, y) | x, y \geq 0\}$$

$$\mathbf{u} + \mathbf{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$$

$$k\mathbf{u} = k(u_1, u_2) = (ku_1, ku_2)$$

does not satisfy ax. 5



### Example 4. The vector space $M_{22}$

$$V = M_{22}$$

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} + \begin{pmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{pmatrix} = \begin{pmatrix} u_{11} + v_{11} & u_{12} + v_{12} \\ u_{21} + v_{21} & u_{22} + v_{22} \end{pmatrix}$$

$$k\mathbf{u} = k \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} = \begin{pmatrix} ku_{11} & ku_{12} \\ ku_{21} & ku_{22} \end{pmatrix}$$



**Example 5.**  $V = \mathbb{R}^+$

$$u + v = uv$$

$$k.u = u^k$$



## Example 6. The space of polynomials

- $V_1 = P_2[x] = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$   
 $(a_1x^2 + b_1x + c_1) + (a_2x^2 + b_2x + c_2) = (a_1 + a_2)x^2 + (b_1 + b_2)x + (c_1 + c_2)$   
 $k.P_2[x] = k(ax^2 + bx + c) = (ka)x^2 + (kb)x + kc$



## Example 6. The space of polynomials

- $V_1 = P_2[x] = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$   
 $(a_1x^2 + b_1x + c_1) + (a_2x^2 + b_2x + c_2) = (a_1 + a_2)x^2 + (b_1 + b_2)x + (c_1 + c_2)$   
 $k.P_2[x] = k(ax^2 + bx + c) = (ka)x^2 + (kb)x + kc$
- $V_2 = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}, a \neq 0\}$



## Example 6. The space of polynomials

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 $(a_1x^2 + b_1x + c_1) + (a_2x^2 + b_2x + c_2) = (a_1 + a_2)x^2 + (b_1 + b_2)x + (c_1 + c_2)$   
 $k.P_2[x] = k(ax^2 + bx + c) = (ka)x^2 + (kb)x + kc$
- $V_2 = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}, a \neq 0\}$  NOT a vector space (not closed under addition)



## Linear combination

Let  $M = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  be a subset of  $V$ . A vector  $\mathbf{v} \in V$  is called a linear combination of  $M$  if  $\exists c_1, \dots, c_p \in K$  such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p$$

Vector  $\mathbf{b}$  is a linear combination of the columns of a matrix  $A$  if and only if the equation  $A\mathbf{x} = \mathbf{b}$  has a solution.



## Linear independence

An indexed set of vectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  in  $V$  is said to be **linearly independent** if the vector equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_p \mathbf{v}_p = \mathbf{0}$$

has *only* the trivial solution,  $c_1 = 0, \dots, c_p = 0$ .





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has *only* the trivial solution,  $c_1 = 0, \dots, c_p = 0$ .

The columns of a matrix  $A$  are linearly independent if and only if the equation  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.



## Linear dependence

The set  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  in  $V$  is said to be **linearly dependent** if the vector equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_p \mathbf{v}_p = \mathbf{0}$$

has a nontrivial solution. Then the equation is called a **linear dependence relation** among  $\mathbf{v}_1, \dots, \mathbf{v}_p$ .



## Linear dependence

The set  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  in  $V$  is said to be **linearly dependent** if the vector equation

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has a nontrivial solution. Then the equation is called a **linear dependence relation** among  $\mathbf{v}_1, \dots, \mathbf{v}_p$ .

The columns of a matrix  $A$  are linearly dependent if and only if the equation  $A\mathbf{x} = \mathbf{0}$  has nontrivial solutions.



## Example

Let the set of vectors  $M = \{(1, 2, 3), (4, 5, 6), (2, 1, 0)\}$ . Determine if  $M$  is linearly independent.



## Example

Let  $M = \{(1, 1, 1), (2, 1, 3), (1, 2, 0)\}$ .

- Is  $M$  linearly independent?
- Is vector  $\mathbf{v} = (2, -1, 3)$  a linear combination of  $M$ ?



## Example

Let  $M = \{1 + t + t^2, 2 + t + 3t^2, 1 + 2t\}$ .

- Is  $M$  linearly independent?
- Is  $P_2[t] = 2 - t + 3t^2$  a linear combination of  $M$ ?



## Example

Let  $M = \left\{ \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \right\}$ .

- Is  $M$  linearly independent?
- Is  $\begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$  a linear combination of  $M$ ?

• **Solution** Check whether

$$a \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + b \begin{pmatrix} 2 & 1 \\ 3 & 0 \end{pmatrix} + c \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a + 2b + c & a + b + 2c \\ a + 3b & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (1)$$

$$\text{implies } a = b = c = 0. \text{ Eq(1)} \Leftrightarrow \begin{cases} a + 2b + c = 0 \\ a + b + 2c = 0 \\ a + 3b = 0 \end{cases} \Leftrightarrow \begin{cases} a = -3t \\ b = t \\ c = t \end{cases}, t \in \mathbb{R}$$

Thus there is non-trivial solution for eq(1). Therefore,  $M$  is linearly dependent.



## Example (cont.)

Let  $M = \left\{ \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \right\}$ .

- Is  $M$  linearly independent?
- Is  $\begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$  a linear combination of  $M$ ?

- **Solution** Check whether there exist  $a, b, c$  such that

$$a \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + b \begin{pmatrix} 2 & 1 \\ 3 & 0 \end{pmatrix} + c \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a + 2b + c & a + b + 2c \\ a + 3b & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{cases} a + 2b + c = 2 \\ a + b + 2c = -1 \\ a + 3b = 3 \end{cases} \Leftrightarrow \left( \begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & -1 \\ 1 & 3 & 0 & 3 \end{array} \right) \sim \left( \begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{array} \right).$$

The system has no solution. Thus, the matrix is NOT a linear combination of  $M$ .





## Example

Determine if the columns of the matrix  $A = \begin{pmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{pmatrix}$  are linearly independent.

**Hint** Check if  $\det(A) \neq 0$ , **or** check if  $a \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix} + b \begin{pmatrix} 1 \\ 2 \\ 8 \end{pmatrix} + c \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix} = 0$  has only the trivial solution  $a = b = c = 0$ .



## Example

Let  $M = \{x, y, z\}$  be a linearly independent set in a vector space  $V$ . Show that  $M_1 = \{x + y + 2z, 2x + 3y + z, 3x + 4y + z\}$  is linearly independent.



## Theorem (Characterization of Linearly Dependent Sets)

An indexed set  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  of two or more vectors, with  $\mathbf{v}_1 \neq \mathbf{0}$ , is linearly dependent if and only if some  $\mathbf{v}_j$  (with  $j > 1$ ) is a linear combination of the preceding vectors,  $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$ .

## Theorem

Let  $M = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$  and  $N = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ . If every  $\mathbf{u}_k$  is a linear combination of  $M$  and  $n > m$ , then  $N$  is linearly dependent.



## Properties

- If  $M$  is linearly dependent and  $N \supset M$ , then  $N$  is linearly dependent.
- If  $M$  is linearly independent and  $N \subset M$ , then  $N$  is linearly independent.
- Any set  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\} \subset \mathbb{R}^n$  is linearly dependent if  $p > n$ .
- Any set  $M$  containing the zero vector is linearly dependent.



## Example

Determine if the given set is linearly dependent

**a**  $\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix}$

**b**  $\begin{pmatrix} 1 \\ 7 \\ 6 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 9 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 5 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \\ 8 \end{pmatrix}$

**c**  $\begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 8 \end{pmatrix}$

**d**  $\begin{pmatrix} -2 \\ 4 \\ 6 \\ 10 \end{pmatrix}, \begin{pmatrix} 3 \\ -6 \\ -9 \\ 15 \end{pmatrix}$

## Example

Determine if the given set is linearly dependent

**a**  $M = \{(1, 1, 1), (2, 1, 3), (1, 2, 0)\}$

**b**  $M = \{x^2 + x + 1, 2x^2 + 3x + 2, 2x + 1\}$

**c**  $\left\{ \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \begin{pmatrix} 3 & 4 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix} \right\}$

**Solution c** Check if there exist scalars  $a, b, c, d$ , not all zero, such that

$$a \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + b \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} + c \begin{pmatrix} 3 & 4 \\ 0 & 1 \end{pmatrix} + d \begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} a + 2b + 3c + d & a + b + 4c + 3d \\ a + b - d & -b + c + 2d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \begin{cases} a + 2b + 3c + d = 0 \\ a + b + 4c + 3d = 0 \\ a + b - d = 0 \\ -b + c + 2d = 0 \end{cases} \Leftrightarrow \begin{cases} a = 0 \\ b = 1 \\ c = -1 \\ d = 1 \end{cases}.$$

Thus the set is linearly dependent.

## Example

Determine all value(s)  $m$ , such that the following set is linearly dependent  
 $M = \{(1, 1, 0); (1, 2, 1), (m, 0, 1)\}$

**Solution** Let  $A = \begin{pmatrix} 1 & 1 & m \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ . The set is linearly dependent if and only if  
 $\det(A) = 0 \Leftrightarrow m + 1 = 0 \Leftrightarrow m = -1$ .



## Rank of Vectors

The rank of  $M \subset V$  is  $k$  if there exists  $k$  linearly independent vectors in  $M$  and any subset of  $M$  containing more than  $k$  vectors is linearly dependent.

### Remark

The rank of  $M$  is the maximum number of independent vectors from  $M$ .





## Example

Find a rank of following set of vectors

$$M = \{(1, 1, 1, 0); (1, 2, 1, 1), (2, 3, 2, 1), (1, 3, 1, 2)\}$$



## Example

Let  $M = \{x, y\} \subset V$  be linearly independent. Identify the rank of the following subset

**a**  $M_1 = \{2x, 3y\}$

**b**  $M_2 = \{x, y, 2x + 3y\}$

**c**  $M_3 = \{x, y, 2x + 3y, 0\}$



## Definition

A set  $M$  is called a **spanning set** for vector space  $V$  if any vector in  $V$  is a linear combination of  $M$ . Denote  $V = \text{Span}(M)$  or  $V = \langle M \rangle$ .



## Example

Determine whether each following set is a spanning set of  $V$ .

**a**  $M = \{(1, 1, 1); (1, 2, 1); (2, 3, 1)\}$ ,  $V = \mathbb{R}^3$ .

**b**  $M = \{(1, 1, 1); (2, 3, 1); (3, 4, 0)\}$ ,  $V = \mathbb{R}^3$ .

**c**  $\{x^2 + x + 1; 2x^2 + 3x + 1; x^2 + 2x\}$ ,  $V = \mathbb{R}_2[x]$ .



## Definition

The **column space** of an  $m \times n$  matrix  $A$ , written as  $Col(A)$ , is the set of all linear combinations of the columns of  $A$ . If  $A = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ , then

$$Col(A) = \{\mathbf{b} : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\} = Span\{\mathbf{v}_1, \dots, \mathbf{v}_n\}.$$



## Example

Let  $A = \begin{pmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} 3 \\ 3 \\ -4 \end{pmatrix}$ . Determine whether  $\mathbf{b}$  is in the column space of  $A$ .

The vector  $\mathbf{b}$  is a linear combination of the columns of  $A$  if and only if  $\mathbf{b}$  can be written as  $A\mathbf{x}$  for some  $\mathbf{x}$ , that is, if and only if the equation  $A\mathbf{x} = \mathbf{b}$  has a solution.

$(A|\mathbf{b}) \sim \left( \begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -2 & -6 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right)$ ,  $\text{rank}(A) = \text{rank}(A|\mathbf{b}) \rightarrow$  The system is consistent (has infinitely many solutions). Thus  $\mathbf{b} \in \text{Col}(A)$ .



## Example

Find a matrix  $A$  such that  $W = \text{Col}(A)$ .  $W = \left\{ \begin{pmatrix} 6a - b \\ a + b \\ -7a \end{pmatrix} : a, b \in \mathbb{R} \right\}$

First, write  $W$  as a set of linear combinations.

$$W = \left\{ a \begin{pmatrix} 6 \\ 1 \\ -7 \end{pmatrix} + b \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} : a, b \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{pmatrix} 6 \\ 1 \\ -7 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\}$$

Second, use the vectors in the spanning set as the columns of  $A$ . Let  $A = \begin{pmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{pmatrix}$ .

Then  $W = \text{Col}(A)$ , as desired.



## Definition

The set of all linear combinations of the row vectors is called the **row space** of  $A$  and is denoted by  $Row(A)$ .

**Example.** Let  $A = \begin{pmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{pmatrix}$  The row space of  $A$  is the subspace of  $\mathbb{R}^5$  spanned by  $\mathbf{r}_1; \mathbf{r}_2; \mathbf{r}_3; \mathbf{r}_4$ . That is,  $Row(A) = Span\{\mathbf{r}_1; \mathbf{r}_2; \mathbf{r}_3; \mathbf{r}_4\}$ .





## Theorem

Let  $A$  be an  $m \times n$  matrix over  $K$  ( $K$  is  $\mathbb{R}$  or  $\mathbb{C}$ ). Then

- $\text{rank}(A) = \text{rank}(\text{Row}(A))$
- $\text{rank}(A) = \text{rank}(\text{Col}(A))$

## Proposition

Let  $M$  be a set containing  $m$  vectors.

- If  $\text{rank}(A) = m$  then  $M$  is independent.
- if  $\text{rank}(A) < m$  then  $M$  is dependent.



## Example

Find the rank of the set  $M = \{(1, 1, 1, 0); (1, 1, -1, 1); (2, 3, 1, 1); (3, 4, 0, 2)\}$



## Example

Let  $A = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 3 & 1 & 0 & 5 \\ -2 & 4 & 1 & 6 \end{pmatrix}$ . Find the set of row vectors and column vectors of  $A$ . Then identify the corresponding ranks.



## Definition

Let  $V$  be vector space.  $B = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$  is a **basic** for  $V$  if

- ①  $B$  is linearly independent.
  - ②  $B$  is a spanning set of  $V$ .
- 
- A basis is a spanning set that is as small as possible.
  - A basis is an independent set that is as many as possible.
  - If  $B$  is a finite set then  $V$  is said to be finite-dimensional.
  - If  $B$  is an infinite set, then  $V$  is said to be infinite-dimensional.



## Theorem

If a vector space  $V$  has a basis of  $n$  vectors, then every basis of  $V$  must consist of exactly  $n$  vectors.  $\dim(V) = n$ .

- ① bases: infinitely many
- ② dimension: unique



## Theorem (The Spanning Set Theorem)

Let  $S = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$  be a set in  $V$ , and let  $H = \text{Span}(\mathbf{x}_1, \dots, \mathbf{x}_n)$ .

- 1 If  $\mathbf{x}_k$  is a linear combination of the remaining vectors in  $S$ , then removing  $\mathbf{x}_k$  from  $S$  does not matter.
- 2 If  $H \neq \{0\}$ , then some subset of  $S$  is a basis of  $H$ .



## Example

- $\dim \mathbb{R}^n = n, \dim \mathbb{R}_2[x] = 3.$
- Find the dimension of the column space of the following matrices.

$$\begin{pmatrix} 1 & 2 & -4 & 3 & -2 & 6 & 0 \\ 0 & 0 & 0 & 1 & 0 & -3 & 7 \\ 0 & 0 & 0 & 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}$$

**Hint** The dimension of the column space of a matrix is equal to the number of its pivot columns.



## Example

Find the dimensions of the vector space spanned by the given vectors:

$$\left\{ \begin{pmatrix} 1 \\ -5 \end{pmatrix}, \begin{pmatrix} -2 \\ 10 \end{pmatrix}, \begin{pmatrix} -3 \\ 15 \end{pmatrix} \right\}$$





## Example

Find the dimensions of the vector space spanned by the given vectors:

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} \right\}$$



## Example

Find the dimension of the vector space.  $H = \left\{ \begin{pmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$



## Theorem

Let  $M = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$  be a basis of a finite-dimensional vector space  $V$ . Then

- Any set of more than  $n$  vectors is linearly dependent.
- Any set of less than  $n$  vectors cannot span  $V$ .



## Theorem

Suppose that  $\dim(V) = n$ . Then

- Any independent set in  $V$  containing exactly  $n$  vectors is a basis of  $V$ .
- Any spanning set in  $V$  containing exactly  $n$  vectors is a basis of  $V$ .

## Theorem

If a vector space  $V$  has a basis of  $n$  vectors, then every basis of  $V$  must consist of exactly  $n$  vectors.



## Example

- Let  $\mathbf{e}_i$  be the columns of  $I_n$ . The set  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  is called the standard basis for  $\mathbb{R}^n$ .
- Let  $S = \{1, t, t^2, \dots, t^n\}$ . The set  $S$  is a standard basis for  $\mathbb{R}_n[x]$ .



## Example

Determine if  $\{(3, 0, -6), (-4, 1, 7), (-2, 1, 5)\}$  is a basis for  $\mathbb{R}_3$ .

**Hint** Check if the three vectors are linearly independent.



## Theorem

The pivot columns of a matrix  $A$  form a basis for  $\text{Col}(A)$ .

## Example

Find a basis for  $\text{Col}(B)$ , where

$$B = \begin{pmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$



## Lemma

If  $A$  is row equivalent to  $B$ , then the columns of  $A$  have the same linear independence/dependence relationships as the columns of  $B$  do.

## Example

Assume that  $A$  is row equivalent to  $B$  with

$$A = \begin{pmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{pmatrix}, B = \begin{pmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \text{ Find a basis for } \text{Col}(A).$$

**Solution** The columns of  $A$  corresponding to the pivot columns of  $B$  form a basis for  $\text{Col}(A)$ . Thus, a basis for  $\text{Col}(A)$  is  $\{(1, 3, 2, 5), (0, 1, 1, 2), (-1, 5, 2, 8)\}$ .





## Example

Determine whether the following set is a basis for  $\mathbb{R}^3$ .  $M = \{(1, 1, 1), (2, 3, 1), (3, 1, 0)\}$ .

**Hint** Check whether the set  $M$  is linearly independent.



## Example

Determine whether the following set is a basis for  $\mathbb{R}_2[x]$ .

$$M = \{x^2 + x + 1; 2x^2 + x + 1; x^2 + 2x + 2\}.$$

